



Day 20 Notes: Open-Source LLM Landscape

◆ 1. What are Open-Source LLMs?

✔ Definition

Open-source LLMs are **large language models whose weights are publicly available** (fully or partially), allowing developers to:

- Run them locally
- Fine-tune them
- Customize for specific use cases



Open vs Closed Models

Feature	Open-Source Models	Closed Models
Access	Weights available	API only
Customization	Full control	Limited
Cost	One-time infra cost	Pay per usage
Examples	LLaMA, Mistral	GPT-4, Claude



Key Insight (Interview Point)

Open-source = flexibility + control

Closed models = performance + convenience

◆ 2. Popular Open-Source LLMs

🧠 1. LLaMA (Meta)

- Variants: 7B, 13B, 70B
- Strong general-purpose model
- Widely used base for fine-tuning

📌 Use case:

- Custom enterprise applications
- Fine-tuning pipelines

⚡ 2. Mistral / Mixtral

- Mistral: Efficient dense model
- Mixtral: Mixture-of-Experts (MoE)

📌 Key Advantage:

- High performance with **low compute cost**

📌 Use case:

- Production systems (fast + cheap)

🦅 3. Falcon (TII)

- Good performance for its size
- Open and permissive license

📌 Use case:

- Research and experimentation

💎 4. Gemma (Google)

- Lightweight, efficient models
- Designed for developers

◆ 3. Model Size Understanding

Model Size	Meaning
7B	7 billion parameters
13B	Medium capability
70B	High performance

⚖️ Tradeoff Rule

- Small models → Fast, cheap, less accurate
- Large models → Slow, expensive, more accurate

📌 Interview Line:

“Model selection depends on latency, cost, and accuracy requirements.”

◆ 4. Key Tradeoffs (VERY IMPORTANT)

⚖️ 1. Accuracy vs Cost

- GPT-4 → High accuracy, expensive
- Mistral → Good accuracy, cheaper

⚡ 2. Latency vs Quality

- Smaller model → faster response
- Larger model → better reasoning

🔒 3. Privacy vs Convenience

- Local models → secure, private
- API models → easy but data shared

◆ 5. Where to Run Open LLMs

Local Deployment

1. Ollama

- Easiest way to run LLM locally
- Command-based usage

Example:

Bash

```
ollama run mistral
```

2. LM Studio

- GUI-based interface
- No coding required

Cloud Platforms

1. Hugging Face

- Inference API
- Model hosting

2. Together AI

- Fast inference for open models

3. Replicate

- Simple API-based model usage

◆ 6. Practical Hands-On (VERY IMPORTANT)

Goal:

Run an open-source LLM locally

Steps (Ollama Example)

1. Install Ollama
2. Pull model:

Bash

```
ollama pull mistral
```

3. Run:

Bash

```
ollama run mistral
```

4. Try prompts:

- "Explain transformers in simple terms"
- "Write Python code for binary search"

Compare with ChatGPT:

- Response quality
- Speed
- Cost

◆ 7. When to Use Open vs Closed Models

✅ Use Open-Source When:

- You need **data privacy**
 - You want **fine-tuning**
 - You want **low cost at scale**
 - You need **offline/local deployment**
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✅ Use Closed Models When:

- You need **best performance**
 - You want **quick development**
 - No infra setup required
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◆ 8. Real-World Use Cases

🏢 Enterprise Use

- Internal chatbot (private data)
 - Document analysis (RAG)
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🚀 Startup Use

- Cost-efficient AI products
 - Custom domain models
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💻 Developer Use

- Experimentation
 - Learning LLM internals
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◆ 9. Interview Questions (Must Prepare)

? Q1: Open vs Closed LLM?

👉 Answer:

- Open = customizable, local deployment
- Closed = better performance, API-based

? Q2: Why choose Mistral over GPT-4?

👉 Answer:

- Lower cost
- Faster inference
- Can run locally

? Q3: What is model size impact?

👉 Answer:

- Larger models → better reasoning
- Smaller models → faster, cheaper

? Q4: How do you deploy open LLMs?

👉 Answer:

- Local (Ollama)
- Cloud (HF, Together AI)

◆ 10. Key Summary (Revise This)

- Open LLMs = control + flexibility
- Closed LLMs = performance + simplicity
- Model selection = **cost + latency + accuracy**
- Mistral & LLaMA = most important models
- Ollama = easiest local setup

